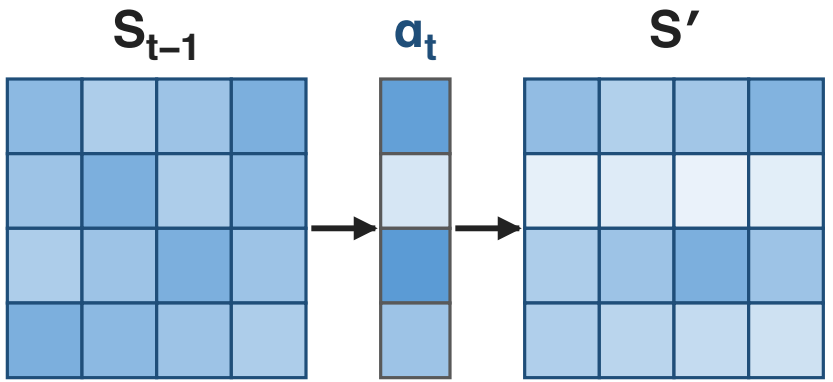
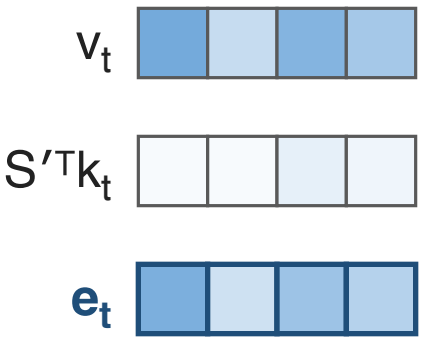
Decay



$$S' = \mathrm{Diag}(\alpha_t)\, S_{t-1}$$

forget per key channel

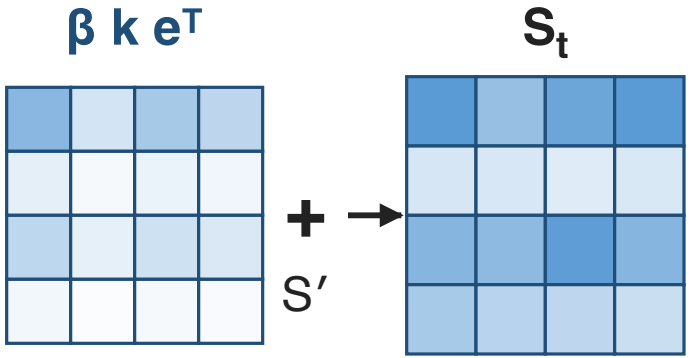
Predict



$$e_t = v_t - S'^T k_t$$

what memory gets wrong

Correct



$$S_t = S' + \beta_t\, k_t\, e_t^T$$

rank-1 write of the error

output $o_t = S_t^T q_t$ · state stays $d_k \times d_v$